Ip addressing & subnetting

Task 1:- ist the IP address classes (A, B, C, D, E) and give one real-world example of a device or service that might use each class. Write your answers in a table format.

IP Class	IP Address Range	Purpose	Real-World Example
Class A	1.0.0.0 – 126.255.255.255	Very large networks	A large company's internal network
Class B	128.0.0.0 – 191.255.255.255	Medium-sized networks	A university campus network
Class C	192.0.0.0 – 223.255.255.255	Small networks	A home Wi-Fi router using `192.168.1.1`
Class D	224.0.0.0 – 239.255.255.255	Multicast communication	Video streaming to multiple devices using multicast
Class E	240.0.0.0 – 255.255.255.255	Experimental and research purposes	Network researchers testing experimental IP protocols

Task 2:- Given the IP address 192.168.10.45 with subnet mask 255.255.255.0, calculate the network address, broadcast address, and the range of valid host addresses. Show your working steps for each calculation.

Given:

- IP Address: 192.168.10.45
- Subnet Mask: 255.255.255.0 (/24)

Step 1: Calculate the Network Address

Rule: IP address AND Subnet Mask = Network Address.

IP Address	**192**	**168**	**10**	**45**
Subnet Mask	255	255	255	0
AND Result	192	168	10	0

Network Address = `192.168.10.0`

Working: The first three octets are unchanged because the subnet mask is 255. The last octet becomes 0 because the subnet mask is 0.

Step 2: Calculate the Broadcast Address

Rule: Set all host bits to 1.

- Subnet mask: `255.255.255.0`
- Host portion: Last 8 bits
- All host bits set to 1 = `11111111` = 255

Therefore:

Broadcast Address = `192.168.10.255`

Step 3: Calculate the Valid Host Address Range

The valid host range starts one address after the network address and ends one address before the broadcast address.

- First valid host: `192.168.10.0 + 1` = `192.168.10.1`
- Last valid host: `192.168.10.255 – 1` = `192.168.10.254`

Number of valid hosts:

\[2^8-2=256-2=\boxed{254} \]

We subtract 2 because the network and broadcast addresses cannot be assigned to ordinary hosts.

Final Answer

Item	Result
IP Address	`192.168.10.45`
Subnet Mask	`255.255.255.0 (/24)`
Network Address	192.168.10.0
Broadcast Address	192.168.10.255
First Valid Host	`192.168.10.1`
Last Valid Host	`192.168.10.254`
Total Valid Host Addresses	254

Task 3:- Create a subnetting plan for a small food delivery startup like Zomato that needs 4 separate subnets: one for Delivery Partners, one for Restaurants, one for Customers, and one for Admins. Assign a unique subnet (with network address and subnet mask) for each group using the 10.0.0.0/24 network.

Subnetting Plan for a Food Delivery Startup (Like Zomato)

Given network: 10.0.0.0/24

We need 4 separate subnets for Delivery Partners, Restaurants, Customers, and Admins.

Step 1: Calculate the subnet size

- Original network: `/24`
- Required subnets: 4
- Borrow 2 host bits because \(2^2 = 4\).
- New subnet prefix: `/26`
- Subnet mask: `255.255.255.192`
- Each subnet contains 64 IP addresses, with 62 usable host addresses.

Step 2: Subnetting table

Group	Network Address	Subnet Mask	Usable Host Range	Broadcast Address
Delivery Partners	`10.0.0.0/26`	`255.255.255.192`	`10.0.0.1 – 10.0.0.62`	`10.0.0.63`
Restaurants	`10.0.0.64/26`	`255.255.255.192`	`10.0.0.65 – 10.0.0.126`	`10.0.0.127`
Customers	`10.0.0.128/26`	`255.255.255.192`	`10.0.0.129 – 10.0.0.190`	`10.0.0.191`
Admins	`10.0.0.192/26`	`255.255.255.192`	`10.0.0.193 – 10.0.0.254`	`10.0.0.255`

Step 3: Verify the calculation

- Total IP addresses: \(4 \times 64 = 256\)
- Addresses per subnet: \(2^6 = 64\)
- Usable hosts per subnet: \(64 - 2 = 62\)
- Total usable host addresses: \(4 \times 62 = 248\)

Task 4:- You are given the following scenario: A college WiFi network uses the IP range 172.16.0.0/16. The admin wants to create at least 20 subnets for different buildings. Calculate the new subnet mask and the number of hosts available per subnet. Show your calculation steps and final answers.

Given Information

- Network Address: `172.16.0.0/16`
- Required Subnets: At least 20
- Task: Calculate the new subnet mask and the number of available hosts per subnet.

Step 1: Calculate the bits to borrow

We need at least 20 subnets.

The formula is:

$$ 2^n \geq \text{Required Subnets} $$

- $2^4 = 16$ — not enough.
- $2^5 = 32$ — sufficient.

Therefore, borrow 5 bits from the host portion.

Step 2: Calculate the new subnet mask

Original prefix: `/16`

Borrowed bits: `5`

$$ \text{New Prefix}=16+5= /21 $$

The subnet mask for `/21` is:

```
255.255.248.0
```

Step 3: Calculate the number of hosts per subnet

IPv4 addresses contain 32 bits.

$$ \text{Host bits}=32-21=11 $$

Total addresses per subnet:

$$ 2^{11}=2048 $$

Usable host addresses:

$$ 2^{11}-2=2048-2=\boxed{2046} $$

We subtract 2 for the network address and broadcast address.

Final Answer

Item	Calculation	Answer
Original Network	Given	`172.16.0.0/16`
Required Subnets	Given	At least 20
Bits Borrowed	$2^5=32$	5 bits
Total Subnets	2^5	32 subnets
New Prefix	$16+5$	/21
New Subnet Mask	`/21` converted to decimal	255.255.248.0
Host Bits Remaining	$32-21$	11 bits
Total Addresses per Subnet	2^{11}	2048
Usable Hosts per Subnet	$2048-2$	2046 hosts

Priyansh Bhaliya

Task 5:- A friend sends you a message asking why their device with IP 192.168.1.300 is not connecting to the network. Write a short explanation identifying the error and what a valid IP address in that subnet should look like.

The IP address `192.168.1.300` is invalid because each octet in an IPv4 address must be between 0 and 255. Here, `300` exceeds the maximum allowed value.

A valid IP address in the same `/24` subnet (`255.255.255.0`) would be `192.168.1.100`, with a typical valid host range of `192.168.1.1` to `192.168.1.254` (assuming the network and broadcast addresses are `.0` and `.255`).

Two Ways to Connect a Gaming PC to the Internet

Situation: I have a Wi-Fi router, but my gaming PC only has an Ethernet port and is located far away from the router.

1. Wi-Fi Extender with an Ethernet Port



How it works:

- Place the Wi-Fi extender between the router and the gaming PC, where it receives a good Wi-Fi signal.
- The extender connects wirelessly to the router.
- Connect the gaming PC to the extender using an Ethernet cable with RJ-45 connectors.
- The PC can then access the internet through the extender and router.

Advantage: No long Ethernet cable is needed between the router and PC.

2. Powerline Adapter Kit



How it works:

- Connect the first powerline adapter to the router using an Ethernet cable, then plug it into a wall socket.
- Plug the second adapter into a wall socket near the gaming PC.
- Connect the second adapter to the PC using another Ethernet cable.
- The adapters carry network data over the building's electrical wiring.

Advantage: Provides a wired-style connection without running a long Ethernet cable through the house. Actual performance depends on the electrical wiring and circuit layout.

Comparison

Feature	Wi-Fi extender	Powerline adapter
Connection to router	Wireless	Ethernet
Connection to PC	Ethernet	Ethernet
Main requirement	Strong Wi-Fi signal	Suitable electrical wiring
Gaming performance	Can vary with Wi-Fi interference	Can vary with wiring quality and electrical noise